AI ENGINEERING

Autonomous AI Agents & Multi-Agent Pipeline Blueprint

State Machines, Tool Calling, Vector Memory & Production Failover Architectures

EXECUTIVE PLAYBOOK ARCHITECTURE & CONTENTS:

- LangGraph & State Machine Design Patterns
- Deterministic Tool Calling & JSON Schema Enforcers
- RAG Vector Chunking & Hybrid Search Strategies
- Production Rate Limit & 429 Graceful Fallbacks

BIZTOOLS EXECUTIVE SUITE • COMMERCIAL EDITION

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Agent Orchestration & Deterministic Tool Schemas

CHAPTER 1 OF 2 • Autonomous AI Agents & Multi-Agent Pipeline Blueprint

Building Resilient Multi-Agent Pipelines

Coordinate specialized worker agents with clear boundaries, shared state dictionaries, and automatic schema validation.

ARCHITECTURE BLUEPRINT: MULTI-AGENT HUB & SPOKE

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Design a 4-agent autonomous pipeline: 1) Researcher Agent, 2) Synthesizer Agent, 3) Reviewer/
Critic Agent, 4) Formatter Agent. Detail input/output schemas, error boundaries, and consensus
voting mechanisms.
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CRITICAL EXECUTION CHECKPOINTS & METRICS:

- [ ] Specify strict JSON Schema for tool calls
- [ ] Implement timeout limits (60s max per step)
- [ ] Include human-in-the-loop escalation gate

Surviving API Downtime and 429 Rate Limits

Production-ready agents must never crash when external LLM providers return rate limits or intermittent outages.

IMPLEMENTATION PATTERN: EXPONENTIAL BACKOFF & LOCAL CACHING

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Detail the architecture for automatic provider failover (Gemini -> Claude -> OpenAI) with
exponential jitter backoff, cached prompt templates, and local rule-based fallback generation.
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CRITICAL EXECUTION CHECKPOINTS & METRICS:

- [ ] Set initial retry backoff to 2s, 4s, 8s
- [ ] Cache static curated assets on disk
- [ ] Log structured telemetry to observability tools